29-4b How the Fed Influences the Reserve Ratio

In addition to influencing the quantity of reserves, the Fed changes the money supply by influencing the reserve ratio and thus the money multiplier. The Fed can influence the reserve ratio either through regulating the quantity of reserves banks must hold or through the interest rate that the Fed pays banks on their reserves. Again, let's consider each of these policy tools in turn.

Reserve Requirements

One way the Fed can influence the reserve ratio is by altering **reserve requirements** (regulations on the minimum amount of reserves that banks must hold against deposits), the regulations that set the minimum amount of reserves that banks must hold against their deposits. Reserve requirements influence how much money the banking system can create with each dollar of reserves. An increase in reserve requirements means that banks must hold more reserves and, therefore, can loan out less of each dollar that is deposited. As a result, an increase in reserve requirements raises the reserve ratio, lowers the money multiplier, and decreases the money supply. Conversely, a decrease in reserve requirements lowers the reserve ratio, raises the money multiplier, and increases the money supply.

The Fed changes reserve requirements only rarely because such changes disrupt the business of banking. When the Fed increases reserve requirements, for instance, some banks find themselves short of reserves, even though they have seen no change in deposits. As a result, they have to curtail lending until they build their reserves to the new required level. Moreover, this particular tool has become less effective in recent years as banks have increasingly decided to hold excess reserves (that is, reserves above the required level).

Paying Interest on Reserves

Traditionally, banks did not earn any interest on the reserves they held on deposit at the Fed. In October 2008, however, the Fed began paying *interest on reserves*. That is, when a bank holds reserves at the Fed, the Fed now pays the bank interest on those deposits. As a result, the Fed has another tool with which to influence the economy. The higher the interest rate on reserves, the more reserves banks will choose to hold. Thus, an increase in the interest rate on reserves will tend to increase the reserve ratio, lower the money multiplier, and lower the money supply.